Task #321 v0.2 — Information Completeness Lemma A.1 substantive work: 14-operator generating set verification

Lyra (Claude Opus 4.7) [Casey 'work the board' directive Sunday 2026-05-24]

2026-05-24 Sunday EDT (~13:30 EDT actual via date)

Task #321 v0.2 — Lemma A.1 Substantive Work

1. Lemma A.1 (restated from v0.1)

Lemma InfoCompleteness-A.1: A_sub is generated by the 14-operator zoo on Bergman H^2(D_IV^5), each generator D_IV^5-derivable from BST primary integers.

2. The 14 generators with substrate-derivation theorems

Per Vol 0 Ch 7 v0.6 + Saturday work + Friday SP-31 #278 STRUCTURALLY VERIFIED batch:

#	Operator	Substrate-derivation theorem	Tier
1	Position x̂_i (Hua-coord)	T2419	STRUCTURALLY VERIFIED
2	Momentum p̂_i (Wirtinger)	T2422	STRUCTURALLY VERIFIED
3	Spin Ŝ_i (SO(5)×SO(2) K-type)	T2421	STRUCTURALLY VERIFIED
4	Angular momentum L̂_i	T2425	STRUCTURALLY VERIFIED
5	Substrate Hamiltonian Ĥ_sub	Elie K52a Session 7+	multi-year pending
6	Bell-CHSH B̂	T2399 + Calibration #17 rank-1 projector	RATIFIED Tuesday May 19

#	Operator	Substrate-derivation theorem	Tier
7	Time reversal T̂	T2433	STRUCTURALLY VERIFIED
8	Charge conjugation Ĉ	T2434	STRUCTURALLY VERIFIED
9	Parity P̂_op	T2472 (Möbius involution + Pin(2) Z_2 lift)	STRUCTURALLY VERIFIED
10	Number N̂	CANDIDATE per Vol 0 Ch 7 v0.6	pending
11	Chirality $\hat{\gamma}^5$	T2471 (SO(2) spinor half-weight)	STRUCTURALLY VERIFIED
12	Electric charge Q̂	T2470 (SO(2) weight operator)	STRUCTURALLY VERIFIED
13	Color Ĉ_3	substrate-mechanism via N_c BST primary	STRUCTURALLY VERIFIED
14	Weak isospin Î_3	substrate-mechanism via Wallach K-type	STRUCTURALLY VERIFIED

3. Lemma A.1 verification at FRAMEWORK level

3.1 Finite count (trivial)

The 14-operator zoo is finite: $14 < \infty$. VERIFIED trivially.

3.2 D_IV^5-derivability per generator

Each of 14 generators has a substrate-derivation theorem listed in Section 2. 11 of 14 are STRUCTURALLY VERIFIED (per Cal #66 tier); 1 RATIFIED (Bell-CHSH B̂ via T2399 RATIFIED Tuesday); 1 multi-year pending (Ĥ_sub via Elie K52a); 1 CANDIDATE pending (N̂).

STATUS: 11+1 = 12/14 substrate-derivation theorems at STRUCTURALLY VERIFIED or RATIFIED tier; 2/14 pending (Ĥ_sub + N̂).

Honest scope: Lemma A.1 requires ALL 14 generators substrate-derived. Currently 12/14 honestly satisfy; 2/14 await closure (Ĥ_sub multi-year via Elie K52a; N̂ multi-month via Vol 0 Ch 7 v0.7+).

3.3 The "generating set" claim (harder)

The non-trivial Lemma A.1 content: does the *-algebra closure of these 14 generators span the Standard Model operator basis?

Standard Model operator basis: - Pauli matrices σ_i ($i = 1, 2, 3$) for SU(2) electroweak - Dirac γ matrices γ_μ ($\mu = 0, 1, 2, 3$) for spin - Gell-Mann matrices λ_a ($a = 1, \ldots, 8$) for SU(3) color - Gauge field strength tensors $F^a_{\mu\nu}$ for SU(3)×SU(2)×U(1) - Higgs sector operators - Yukawa coupling structure operators

Mapping to A_{sub} generators: - Pauli $\sigma_i \leftrightarrow \hat{I}_3$ + extensions (#14) via SU(2) electroweak embedding in K - Dirac $\gamma_\mu \leftrightarrow$ position $\hat{x}i + \hat{S}_i$ *combinations via Clifford algebra inherited from Spin(5,2) - Gell-Mann* $\lambda_a \leftrightarrow \hat{C}_3$ *(#13) via SU(3) color embedding from* N_c *structure - Gauge fields* $F^a\{\mu\nu\} \leftrightarrow$ generated by commutators of K-type operators with momentum - Higgs $\leftrightarrow$ generated by $\hat{Q} + \hat{C}_3 + \hat{I}_3$ closure (Higgs sector operators are gauge-symmetry-breaking polynomials) - Yukawa $\leftrightarrow$ generated by $\hat{S}_i + \hat{C}_3 + \hat{Q}$ closure (Yukawa structure is mass-coupling)

Status of mapping verification: - All 6 SM operator categories have plausible mapping to A_{sub} generators - VERIFICATION of mapping at theorem-grade rigor requires: - Explicit Lie algebra closure check (multi-month) - Verification that A_{sub} *-algebra spans full SM operator basis without gaps - Identification of any SM operator NOT expressible in A_{sub} (would be counter-example)

Honest scope: §3.3 mapping is FRAMEWORK-level argument; theorem-grade requires v0.3+ multi-month Lie-algebra closure verification + explicit SM operator basis enumeration.

4. v0.2 status

What's established (v0.2 additions to v0.1): - §2 14-operator zoo enumeration with substrate-derivation theorem per generator + current tier - §3.1 finite count trivially verified - §3.2 D_{IV}^5-derivability per generator: 12/14 STRUCTURALLY VERIFIED or RATIFIED; 2/14 pending - §3.3 generating set claim at FRAMEWORK level with mapping sketch to SM operator basis

What's NOT established (v0.3+ multi-month): - Generator #5 $\hat{H}_{sub}$ explicit closure (Elie K52a Session 7+ multi-year) - Generator #10 $\hat{N}$ CANDIDATE closure (Vol 0 Ch 7 v0.7+ multi-month) - Lie algebra closure check rigorous (multi-month) - SM operator basis enumeration explicit (multi-month) - Lemma A.1 theorem-grade rigor (multi-month per generator + multi-year for full algebra closure)

Confidence at v0.2: ~40% probability of clean rigorous Lemma A.1 proof within 6-12 months; multi-year pending $\hat{H}_{sub} + \hat{N}$ closure.

Per Calibration #27 STANDING discipline applied: §3.3 mapping is sketch-grade, NOT claimed as theorem-grade rigor; the SM operator basis enumeration must be verified WITHOUT presupposing that A_{sub} spans it.

5. Coordination

Cal: Lemma A.1 FRAMEWORK level pending Cal cold-read with Calibration #27 STANDING applied.

Keeper: K-audit pre-stage for Lemma A.1 at FRAMEWORK tier per Cal #66 (NOT STRUCTURALLY VERIFIED or higher until §3.3 mapping rigorous closure).

Elie: K52a Session 7+ multi-year H_sub closure provides Generator #5 explicit form needed for Lemma A.1 full closure.

Grace: catalog entries for 14-generator substrate-derivation theorem cross-references; SM operator basis catalog for §3.3 verification.

6. v0.2 → v0.3 path

- Generator #10 N̂ CANDIDATE closure (Vol 0 Ch 7 v0.7+ multi-month)
- Generator #5 Ĥ_sub multi-year per K52a
- §3.3 Lie algebra closure rigorous check
- SM operator basis enumeration explicit
- Route B fallback if Route A stalls (per v0.1 §4)

— Lyra, Task #321 v0.2 Lemma A.1 substantive work Sunday 2026-05-24 ~13:30 EDT per Casey "work the board" directive + Calibration #27 STANDING discipline applied. FRAMEWORK level only; theorem-grade rigor multi-month per dependencies.